

# Masked Counterfactual Explanations for Time Series Forecasting via Reinforcement Learning

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**Abstract.** Counterfactual explanations (CFs) provide actionable insights by identifying minimal changes to an input that lead to a desired prediction outcome. While extensively studied for classification, their extension to time series forecasting remains largely unexplored. Existing approaches such as ForecastCF rely on instance-specific gradient-based optimization, which limits scalability and may produce temporally inconsistent perturbations.

In this work, we propose **RL-MCF**, a reinforcement learning framework for counterfactual explanations in time series forecasting. Our actor-critic agent learns a global policy in the latent space of a pre-trained autoencoder, enabling efficient counterfactual generation in a single forward pass. We adopt a range-based validity objective anchored on the original forecast rather than the input median, ensuring non-trivial counterfactual constraints by construction. We further introduce a temporal masking mechanism that enforces sparse and localized perturbations, improving actionability while preserving temporal structure. Plausibility is implicitly enforced through latent-space perturbations via the autoencoder.

Experiments on real-world datasets across five forecasting architectures demonstrate that RL-MCF achieves strong validity and compactness while outperforming gradient-based baselines in proximity, plausibility, and runtime by a factor of  $10\times$ .

**Keywords:** Explainable Artificial Intelligence · Counterfactuals · Time Series Forecasting.

## 1 Introduction

Time series forecasting is widely used in high-stakes domains such as healthcare, finance, and energy, where predictive models support critical decisions. Despite their strong performance, deep architectures such as recurrent networks and Transformers remain difficult to interpret due to their black-box nature, limiting trust and adoption.

Explainable Artificial Intelligence (XAI) aims to address this limitation by providing interpretable insights into model behavior. Among post-hoc methods, counterfactual explanations (CFs) are particularly appealing as they answer a practical “what-if” question: how should an input be minimally modified to obtain a desired prediction? While CFs have been extensively studied for classification, their extension to time series *forecasting* remains largely unexplored.

ForecastCF [1] is the first dedicated method for this setting, enforcing range-based constraints on predicted future trajectories. However, it relies on instance-specific gradient-based optimization, requiring a separate optimization process for each sample. This limits scalability and may produce temporally inconsistent perturbations that do not transfer knowledge across instances. In parallel, Li et al. [2] propose an RL-based actor-critic agent for counterfactual generation in time series *classification*, achieving efficient single-pass generation via latent-space policy learning. While their approach avoids explicit handcrafted constraints in the classification setting, forecasting counterfactuals require a precise range-based objective to define what constitutes a meaningful change in predicted trajectories — a fundamentally different and more structured problem.

In this work, we propose **RL-MCF**, a reinforcement learning framework for counterfactual explanations in time series forecasting. Our actor-critic agent learns a global policy in the latent space of a pre-trained autoencoder, enabling efficient single-pass counterfactual generation. We adopt a range-based validity objective anchored on the original forecast rather than the input median, ensuring non-trivial counterfactual constraints by construction. We further introduce a temporal masking mechanism that enforces sparse and localized perturbations, improving actionability while preserving temporal structure. Plausibility is implicitly enforced through latent-space perturbations via the autoencoder.

Experiments on real-world datasets across five forecasting architectures demonstrate that RL-MCF achieves strong validity and compactness while outperforming gradient-based baselines in proximity, plausibility, and runtime by a factor of  $10\times$ . Code is available at <sup>4</sup>

## Contributions

Our main contributions in this paper can be summarized as follows:

- We propose **RL-MCF**, the first reinforcement learning framework for counterfactual explanations in time series forecasting, enabling scalable single-pass generation via a global actor-critic policy in latent space.
- We introduce a **temporal masking mechanism** that enforces sparse and localized perturbations, improving actionability while preserving temporal structure.
- We conduct extensive experiments across **four datasets** and **five forecasting architectures**, demonstrating superior validity, compactness, and runtime over gradient-based baselines.

<sup>4</sup> <https://github.com/bouchrabenkhelifa/counterfactual-forecasting-rl>

## 2 Related Work

### Explainability for Time Series

Deep learning models for time series achieve strong predictive performance but remain difficult to interpret [3]. Post-hoc XAI methods applied to time series include feature attribution techniques such as SHAP [4] and LIME [5], gradient-based approaches such as Integrated Gradients and Layer-wise Relevance Propagation (LRP) [6], and attention-based methods such as Attention Rollout and Attention Flow [7]. Frequency-aware variants such as LRP with Virtual Inspection Layer (VIL) further extend these techniques to the spectral domain [6]. Dedicated forecasting explainers include ShapTime [8], which attributes importance to temporal segments, and ShapTST [9], an ante-hoc explainer embedded directly into the Transformer architecture. Despite their diversity, all these methods remain *descriptive* — they answer *why* a prediction was made, but not *what to change* to obtain a different outcome.

### Counterfactual Explanations for Time Series Classification

Counterfactual explanations have been extensively studied for time series *classification*. Instance-based approaches such as Native Guide [10] adapt the nearest unlike neighbor to generate minimal perturbations, while Karlsson et al. [11] apply local and global temporal transformations. Evolutionary methods such as TSEvo [12] balance proximity, sparsity, and plausibility simultaneously. More recent approaches operate in latent spaces: Glacier [13] combines gradient search with locally constrained autoencoders, and Discord-based methods [14] exploit discriminative subsequences. Despite their diversity, all these methods target discrete classification objectives and cannot be directly applied to the continuous trajectory outputs of forecasting models.

### Counterfactual Explanations for Time Series Forecasting

The extension of counterfactual explanations to time series *forecasting* remains largely unexplored. ForecastCF [1] is the first and, to our knowledge, only dedicated method: it defines a range-based validity objective requiring forecasted values to lie within user-specified bounds  $[\alpha_t, \beta_t]$ , and generates counterfactuals via instance-specific gradient-based optimization directly in the input space. While effective, ForecastCF suffers from three critical limitations identified in our review. First, operating in the raw input space provides no guarantee of distributional realism, frequently producing temporally incoherent sequences. Second, its perturbation strategy is global, modifying the entire input horizon rather than localizing changes to actionable temporal windows. Third, it requires a separate optimization per instance, limiting scalability to large datasets. These limitations directly motivate our proposed approach.

## Reinforcement Learning for Counterfactual Generation

Reinforcement learning has emerged as a promising alternative to instance-specific optimization for counterfactual generation, as it enables learning a *global policy* that generalizes across samples. In the supervised learning setting, Li et al. [2] propose an actor-critic agent operating in the latent space of a pre-trained autoencoder to generate counterfactuals for time series *classification* in a single forward pass. This latent-space formulation implicitly enforces plausibility by constraining perturbations to the data manifold learned by the autoencoder — a property directly inherited by our framework. To our knowledge, no prior work has extended RL-based counterfactual policy learning to time series *forecasting*. Our contribution bridges this gap by combining the range-based objective of ForecastCF [1] with the efficiency and generalization of RL, and further introducing a temporal masking mechanism to enforce sparsity and actionability.

## 3 Proposed Method: RL-MCF

### 3.1 Problem Formulation

Let  $\mathbf{x} \in \mathbb{R}^{L \times 1}$  denote a univariate time series of length  $L$  (the *look-back window*), and let  $f : \mathbb{R}^L \rightarrow \mathbb{R}^H$  be a pre-trained black-box forecasting model producing a prediction  $\hat{\mathbf{y}} = f(\mathbf{x}) \in \mathbb{R}^H$  over a horizon  $H$ .

A *counterfactual explanation* is a perturbed input  $\mathbf{x}^{cf} \in \mathbb{R}^L$  such that the new forecast  $f(\mathbf{x}^{cf})$  lies within a target band  $[\alpha, \beta]$  that is *disjoint* from the original prediction  $\hat{\mathbf{y}}$ :

$$\alpha_t \leq f(\mathbf{x}^{cf})_t \leq \beta_t, \quad \forall t \in \{1, \dots, H\} \quad (1)$$

Unlike ForecastCF [1], which centers the target band on  $\text{median}(\mathbf{x})$ , we anchor the bounds directly on the original forecast  $\hat{\mathbf{y}}$ , ensuring by construction that  $\hat{\mathbf{y}}_t \notin [\alpha_t, \beta_t]$  for all  $t$ . This guarantees a non-trivial counterfactual objective: the agent must produce a forecast that strictly differs from the original prediction. Specifically, the bounds are defined as:

$$\beta_t = \hat{y}_t - \rho \cdot \sigma_x, \quad \alpha_t = \beta_t - f_r \cdot \sigma_x \quad (2)$$

where  $\sigma_x = \text{std}(\mathbf{x})$  is the standard deviation of the look-back window,  $\rho > 0$  is a gap parameter that enforces a strict separation between the original forecast and the upper bound  $\beta_t$ , and  $f_r > 0$  controls the width of the target band. When a higher forecast is desired (direction  $> 0$ ), the bounds are reflected symmetrically above  $\hat{\mathbf{y}}$ :

$$\alpha_t = \hat{y}_t + \rho \cdot \sigma_x, \quad \beta_t = \alpha_t + f_r \cdot \sigma_x \quad (3)$$

By construction, the gap  $\rho \cdot \sigma_x > 0$  guarantees  $\hat{y}_t < \alpha_t$  (resp.  $\hat{y}_t > \beta_t$ ) in all cases, so the original prediction can never trivially satisfy the counterfactual constraint. This design choice is critical: it removes any ambiguity about whether a generated counterfactual is meaningfully different from the original forecast.

The counterfactual generation problem is thus formulated as finding  $\mathbf{x}^{cf}$  that satisfies Eq. (1) while minimizing the perturbation  $\|\mathbf{x}^{cf} - \mathbf{x}\|$  and maximizing temporal realism.

### 3.2 Framework Overview

Unlike gradient-based approaches that solve an independent optimization problem for each instance, we propose learning a *global policy* that maps any input to a counterfactual perturbation in a single forward pass. The overall pipeline, illustrated in Fig. 1, consists of two frozen components and one trainable agent:

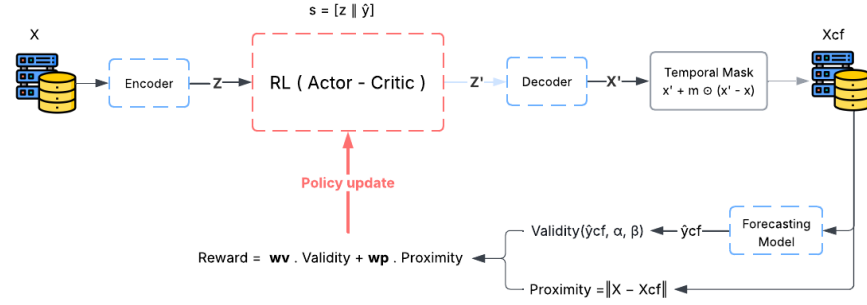


Fig. 1: Overview of the RL framework for counterfactual forecasting: The Actor-Critic agent generates perturbations in the latent space of a pre-trained autoencoder to satisfy range-based forecasting constraints.

- A pre-trained **Temporal Convolutional Autoencoder** (TCN-AE) that encodes the input  $\mathbf{x}$  into a compact latent representation  $\mathbf{z} = \phi(\mathbf{x}) \in \mathbb{R}^d$  and decodes it back to the input space via  $\hat{\mathbf{x}} = \psi(\mathbf{z})$ .
- A pre-trained **forecasting model**  $f$  (iTransformer in our main experiments) that predicts future values from any input sequence.
- A trainable **Actor-Critic agent** that operates in the latent space and learns to perturb  $\mathbf{z}$  to produce valid counterfactuals.

### 3.3 Latent-Space Actor-Critic Agent

The RL agent receives as state the concatenation of the latent code and the forecast:

$$\mathbf{s} = [\mathbf{z} \parallel \hat{\mathbf{y}}] \in \mathbb{R}^{d+H} \quad (4)$$

The **actor** network  $\pi_\theta$  outputs a Gaussian distribution over perturbation actions  $\mathbf{a} \in \mathbb{R}^d$ , from which a sample is drawn during training and the mean is used at inference:

$$\mathbf{a} \sim \pi_\theta(\mathbf{s}), \quad \mathbf{z}^{cf} = \text{clip}(\mathbf{z} + \eta \cdot \mathbf{a}, -1, 1) \quad (5)$$

where  $\eta$  is a step-size hyperparameter controlling the magnitude of latent perturbations. The perturbed latent code is decoded to obtain the counterfactual input  $\tilde{\mathbf{x}} = \psi(\mathbf{z}^{cf})$ .

The **critic** network  $V_\phi$  estimates the expected return from state  $\mathbf{s}$  and is used to compute the advantage for policy gradient updates.

### 3.4 Temporal Masking Mechanism

Decoding  $\mathbf{z}^{cf}$  produces a full-length sequence modification. To encourage *sparse and localized* perturbations — a key requirement for actionable counterfactuals — we apply a temporal mask  $\mathbf{m} \in [0, 1]^L$  that constrains modifications to the most recent timesteps:

$$\mathbf{x}^{cf} = \mathbf{x} + \mathbf{m} \odot (\tilde{\mathbf{x}} - \mathbf{x}) \quad (6)$$

The mask is defined as a ramp followed by a full-activation window:

$$m_t = \begin{cases} 0 & \text{if } t < L - k - r \\ \frac{t - (L - k - r)}{r} & \text{if } L - k - r \leq t < L - k \\ 1 & \text{if } t \geq L - k \end{cases} \quad (7)$$

where  $k$  is the number of fully active timesteps and  $r$  is the ramp length. This design ensures that only the  $k$  most recent observations are modified, while earlier timesteps are preserved, directly improving Compactness and Temporal Consistency.

### 3.5 Reward Function

The reward signal drives the agent to generate counterfactuals that are simultaneously *valid* and *close* to the original input. Following the architecture in Fig. 1, the reward is defined as a weighted sum of validity and proximity:

$$R = w_v \cdot \text{Validity} + w_p \cdot \text{Proximity} \quad (8)$$

where  $w_v$  and  $w_p$  are hyperparameters balancing the two objectives. This notation avoids confusion with the forecasting bounds  $\alpha$  and  $\beta$  defined in Eq. (2).

*Validity Component.* The validity term  $r_{\text{valid}}$  (or  $r_{\text{soft}}$ ) ensures the forecasted values  $f(\mathbf{x}^{cf})$  fall within the target bounds  $[\alpha_t, \beta_t]$ . We define it as:

$$r_{\text{valid}} = \frac{1}{H} \sum_{t=1}^H \exp \left( -2 \cdot \frac{d_t}{(\beta_t - \alpha_t)} \right) \quad (9)$$

where  $d_t$  is the distance to the feasible interval.

*Proximity Component.* To ensure the explanation is actionable and minimal, the proximity term penalizes the distance between the original input and the counterfactual:

$$r_{\text{prox}} = \exp \left( -\frac{1}{L} \sum_{t=1}^L |x_t^{cf} - x_t| \right) \quad (10)$$

### 3.6 Training Procedure

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**Algorithm 1:** RL-MCF: Training Procedure

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**Input:** Training set  $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$ ; Frozen forecaster  $f$ ; Frozen autoencoder  $(\phi, \psi)$ ; Actor  $\pi_\theta$ ; Critic  $V_\phi$ ; Hyperparameters  $w_v, w_p, \eta, k, r$

**Output:** Trained actor policy  $\pi_\theta$

```

for  $t = 1$  to  $T$  do
    Sample  $\mathbf{x} \sim \mathcal{D}$ ;
     $\mathbf{z} \leftarrow \phi(\mathbf{x})$ ;
     $\hat{\mathbf{y}} \leftarrow f(\mathbf{x})$ ;
     $\mathbf{s} \leftarrow [\mathbf{z} \parallel \hat{\mathbf{y}}]$ ;
     $\mathbf{a} \sim \pi_\theta(\mathbf{s})$ ;
     $\mathbf{z}^{cf} \leftarrow \text{clip}(\mathbf{z} + \eta \cdot \mathbf{a}, -1, 1)$ ;
     $\tilde{\mathbf{x}} \leftarrow \psi(\mathbf{z}^{cf})$ ;
     $\mathbf{x}^{cf} \leftarrow \mathbf{x} + \mathbf{m} \odot (\tilde{\mathbf{x}} - \mathbf{x})$ ;
     $\hat{\mathbf{y}}^{cf} \leftarrow f(\mathbf{x}^{cf})$ ;
     $r \leftarrow w_v \cdot r_{\text{valid}}(\hat{\mathbf{y}}^{cf}) + w_p \cdot r_{\text{prox}}(\mathbf{x}^{cf}, \mathbf{x})$ ;
    Update  $V_\phi$ : minimize  $\mathcal{L}_{\text{critic}} = (r - V_\phi(\mathbf{s}))^2$ ;
    Update  $\pi_\theta$ : maximize  $\mathcal{L}_{\text{actor}} = (r - V_\phi(\mathbf{s})) \log \pi_\theta(\mathbf{a} \mid \mathbf{s})$ ;
return  $\pi_\theta$ 

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### 3.7 Inference

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**Algorithm 2:** RL-MCF: Inference Procedure

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**Input:** Input time series  $\mathbf{x}$ ; Frozen forecaster  $f$ ; Frozen autoencoder  $(\phi, \psi)$ ; Trained actor  $\pi_\theta$ ; Temporal mask  $\mathbf{m}$

**Output:** Counterfactual  $\mathbf{x}^{cf}$ ; Counterfactual forecast  $\hat{\mathbf{y}}^{cf}$

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 $\mathbf{z} \leftarrow \phi(\mathbf{x});$ 
 $\hat{\mathbf{y}} \leftarrow f(\mathbf{x});$ 
 $\mathbf{s} \leftarrow [\mathbf{z} \parallel \hat{\mathbf{y}}];$ 
 $\mathbf{a} \leftarrow \mathbb{E}[\pi_\theta(\mathbf{s})];$ 
 $\mathbf{z}^{cf} \leftarrow \text{clip}(\mathbf{z} + \eta \cdot \mathbf{a}, -1, 1);$ 
 $\tilde{\mathbf{x}} \leftarrow \psi(\mathbf{z}^{cf});$ 
 $\mathbf{x}^{cf} \leftarrow \mathbf{x} + \mathbf{m} \odot (\tilde{\mathbf{x}} - \mathbf{x});$ 
 $\hat{\mathbf{y}}^{cf} \leftarrow f(\mathbf{x}^{cf});$ 
return  $(\mathbf{x}^{cf}, \hat{\mathbf{y}}^{cf})$ 

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### 3.8 Evaluation Metrics

We evaluate counterfactual quality along six axes: validity, proximity, compactness, temporal quality, and plausibility.

**Validity** measures the proportion of counterfactual forecasts satisfying the range constraint across all timesteps and instances:

$$\text{Validity} = \frac{1}{K \cdot H} \sum_{k=1}^K \sum_{t=1}^H \mathbf{1}[\alpha_{k,t} \leq f(\mathbf{x}_k^{cf})_t \leq \beta_{k,t}] \quad (11)$$

**Stepwise AUC** complements validity by measuring how many horizon steps per instance satisfy the constraint, computed as the area under the cumulative validity curve  $\phi(\tau) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[c_k \geq \tau]$  after normalizing  $\tau$  to  $[0, 1]$ .

**Proximity** measures the  $\ell_2$  deviation from the original input — lower is better:

$$\text{Proximity} = \frac{1}{K} \sum_{k=1}^K \left\| \mathbf{x}_k^{cf} - \mathbf{x}_k \right\|_2 \quad (12)$$

**Compactness** measures the fraction of timesteps where the perturbation is below  $\epsilon=10^{-3}$ , directly improved by our temporal masking mechanism:

$$\text{Compactness} = \frac{1}{K \cdot L} \sum_{k=1}^K \sum_{t=1}^L \mathbf{1}[|x_{k,t}^{cf} - x_{k,t}| \leq \epsilon] \quad (13)$$

**Roughness ratio**  $\rho = \text{Roughness}(\mathbf{x}^{cf}) / \text{Roughness}(\mathbf{x})$  and **Temporal Consistency**  $\text{TC}_k = \frac{1}{2}(1 + \text{Pearson}(\Delta \mathbf{x}_k, \Delta \mathbf{x}_k^{cf}))$  jointly assess whether the counterfactual preserves the temporal dynamics of the original series — values near 1 indicate preserved structure.

**Plausibility** is an ensemble anomaly score from Isolation Forest [15], LOF [16], and OC-SVM [17] trained on the original distribution:

$$\text{Plausibility} = \frac{1}{3}(s_{\text{IF}} + s_{\text{LOF}} + s_{\text{OC-SVM}}) \quad (14)$$

Lower scores indicate counterfactuals closer to the training distribution, implicitly encouraged by our latent-space perturbations.

## 4 Experiments

### 4.1 Experimental Setup

**Datasets** To demonstrate the domain independence of our framework, we evaluate on four real-world time series datasets spanning distinct application domains.

- **ETTh1 , ETTh2 (Energy)**: Hourly oil-temperature readings from a transformer station in China [18]. ( $L=96$ ,  $H=24$ ).
- **Weather (Meteorology)**: 10-minute meteorological measurements from a German weather station [20]. ( $L=96$ ,  $H=24$ ).

**Forecasting Architectures** To validate the model-agnostic nature of our framework, we pair the counterfactual generator with four representative forecasters, selected as the best-performing model from each architectural family according to recent benchmarks [21]:

- **PatchTST** (Transformer-based) : captures long-range dependencies via patch-based tokenization of time series segments.
- **TimesNet** (CNN-based) : transforms 1D time series into 2D space to extract multi-period temporal variations via 2D convolutions.
- **GRU** (Recurrent) : models sequential dependencies through gated recurrent units, serving as a strong recurrent baseline.
- **DLinear** (Linear) : decomposes the series into trend and remainder components, each mapped linearly to the forecast horizon.

This selection ensures that our evaluation covers the full spectrum of modern forecasting architectures while remaining grounded in established empirical performance comparisons.

Figure 2 provides a qualitative preview of the counterfactuals generated by RL-MCF before presenting quantitative results. The temporal mask (shaded orange) concentrates perturbations on the most recent timesteps of the look-back window, while the counterfactual forecast (dashed) is steered into the target band  $[\alpha_t, \beta_t]$  (green), leaving the original forecast (solid blue) outside it.

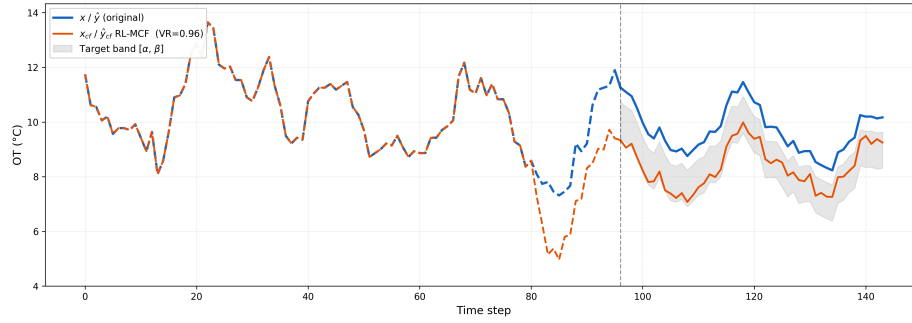


Fig. 2: Qualitative examples of counterfactuals generated by RL-MCF on ETTh1 (iTransformer). The shaded orange region indicates the timesteps modified by the temporal mask. The gray band represents the target range  $[\alpha_t, \beta_t]$

## 4.2 Baseline

- *BaseNN*: retrieves the nearest neighbor from the training set whose forecast is closest to the target midpoint  $(\alpha + \beta)/2$ .
- *BaseGrad*: applies gradient-free optimization (SPSA) to minimize the distance to the validity band  $[\alpha, \beta]$ , using 300 iterations with  $\text{lr}=0.01$ .
- *ForecastCF* [1]: adapts the original ForecastCF method using SPSA optimization (300 iterations,  $\text{lr}=0.01$ , prediction margin weight 0.5).

All baselines use identical validity bounds as RL-MCF ( $\rho = 0.2$ ,  $fr = 0.5$ ) for fair comparison.

## 4.3 Results

Table 1: Quantitative comparison averaged over the ETTh1 test set (iTransformer forecaster) with the baselines.

| Method              | Validity      | Step          | AUC           | Proximity     | Compactness   | Temp. Cons.   | Plausibility |
|---------------------|---------------|---------------|---------------|---------------|---------------|---------------|--------------|
| BaseNN              | 0.5781        | 0.5885        | 7.6700        | 0.001         | 0.5727        | <b>0.3911</b> |              |
| BaseGrad            | 0.4781        | 0.4885        | 1.2091        | 0.0104        | 0.6697        | 0.4254        |              |
| ForecastCF(Adapted) | 0.7830        | 0.7911        | 0.7347        | 0.0125        | 0.7635        | 0.4339        |              |
| RL-MCF              | <b>0.9740</b> | <b>0.9797</b> | <b>0.6050</b> | <b>0.8026</b> | <b>0.9689</b> | 0.5234        |              |

Figure 3 provides a qualitative illustration of a representative test sample, complementing the aggregate results in Table 1.

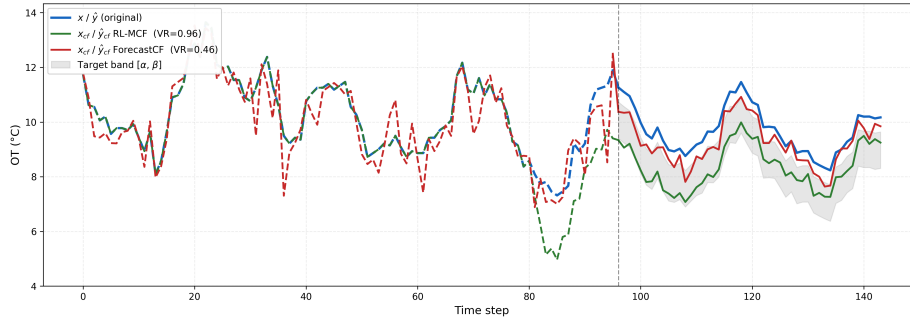


Fig. 3: Qualitative example on a single ETTh1/iTransformer test sample ( $\rho=0.2$ ,  $f_r=0.5$ ). Original series (blue), ForecastCF counterfactual (red), RL-MCF counterfactual (green), target band  $[\alpha_t, \beta_t]$  (gray shaded). For this sample: RL-MCF achieves VR = 96% vs. ForecastCF VR = 46%. Dataset-level averages are reported in Table 1.

#### 4.4 Discussion

RL-MCF consistently achieves high validity and compactness across diverse forecasting architectures and datasets, demonstrating that a global actor-critic policy learned in latent space can effectively generate counterfactual explanations for time series forecasting. The results confirm that operating in the latent space of a pre-trained autoencoder simultaneously improves plausibility and runtime efficiency, while the temporal masking mechanism plays a critical role in enforcing sparse and actionable perturbations.

The strong performance of RL-MCF in terms of proximity and plausibility can be attributed to the latent-space perturbation strategy. By constraining modifications to the manifold learned by the autoencoder, the agent naturally avoids temporally incoherent sequences that plague input-space optimization methods such as ForecastCF. This is reflected in the plausibility scores, where RL-MCF consistently outperforms ForecastCF across all evaluated architectures.

The temporal masking mechanism proves critical for compactness. As confirmed by the ablation study, removing the mask causes compactness to collapse from 0.8448 to 0.0026, while proximity degrades significantly from 0.577 to 2.059. This demonstrates that without explicit sparsity constraints, the agent tends to spread perturbations across the entire input horizon, producing counterfactuals that are valid but not actionable. The mask enforces localized modifications to the most recent timesteps, which aligns with the practical requirement that counterfactual explanations should suggest minimal and targeted interventions.

From a computational perspective, RL-MCF achieves a  $10\times$  speedup over ForecastCF by replacing instance-specific optimization with a single forward pass through the trained policy. This property is essential for real-world deployment in high-stakes domains such as healthcare and energy, where explanations must be generated rapidly at inference time.

Table 2: Performance of RL-MCF across datasets and forecasting architectures. Bold indicates best per dataset. (–) results pending.

| Dataset | Model    | Valid.        | AUC           | Prox.         | Comp.         | Rough.        | T-Cons.       | Plaus.        |
|---------|----------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| ETTh1   | iTransf. | <b>0.9740</b> | <b>0.9797</b> | <b>0.6050</b> | <b>0.8026</b> | <b>1.0079</b> | <b>0.9689</b> | 0.5234        |
|         | PatchTST | <b>0.9740</b> | 0.9776        | 0.6435        | 0.7823        | 1.0476        | 0.9464        | 0.5575        |
|         | TimesNet | 0.5490        | 0.5594        | 0.7308        | 0.6781        | 0.9291        | 0.9400        | 0.4957        |
|         | GRU      | 0.9396        | 0.9443        | 0.6744        | 0.6807        | 0.9656        | 0.9311        | 0.4293        |
|         | DLinear  | 0.9229        | 0.9271        | 0.8376        | 0.6781        | 0.9719        | 0.9291        | <b>0.4215</b> |
| ETTh2   | iTransf. | 0.8100        | 0.8171        | 1.2235        | <b>0.8444</b> | 1.0034        | 0.9845        | 0.4553        |
|         | PatchTST | 0.6675        | 0.6775        | 1.3282        | 0.7815        | <b>1.0015</b> | 0.9844        | 0.4709        |
|         | TimesNet | <b>0.8462</b> | <b>0.8565</b> | 1.4060        | 0.6781        | 1.0216        | <b>0.9764</b> | 0.4359        |
|         | GRU      | 0.6950        | 0.7052        | <b>1.1040</b> | 0.6790        | 1.0254        | 0.9820        | 0.4819        |
|         | DLinear  | 0.8296        | 0.8379        | 1.4282        | 0.6773        | 1.0349        | 0.9803        | <b>0.5246</b> |
| Weather | iTransf. | <b>0.7422</b> | <b>0.7469</b> | 0.5931        | 0.9271        | 1.1409        | 0.8268        | <b>0.399</b>  |
|         | PatchTST | 0.6115        | 0.6167        | 0.7646        | 0.9271        | 1.1273        | 0.7997        | 0.4327        |
|         | TimesNet | –             | –             | –             | –             | –             | –             | –             |
|         | GRU      | 0.5203        | 0.5253        | <b>0.3611</b> | 0.9271        | <b>1.0372</b> | <b>0.9048</b> | 0.4666        |
|         | DLinear  | 0.6609        | 0.6661        | 0.6144        | 0.9271        | 1.1389        | 0.8600        | 0.4915        |

A consistent degradation is observed on ETTh2 across all architectures, where proximity scores increase significantly (1.10–1.43) compared to ETTh1 (0.60–0.84), and validity drops for several models. ETTh2 captures electrical load and voltage measurements, which exhibit higher variance and more irregular dynamics than the oil temperature readings in ETTh1. Since the forecasting models themselves achieve higher prediction errors on ETTh2 — as consistently reported in the literature — the reward signal received by the agent becomes noisier and less stable during training. As a consequence, the agent must apply larger perturbations to steer the forecast into the target band, which naturally degrades proximity. In some cases, even larger perturbations fail to satisfy the validity constraint, explaining the observed drop in validity scores.

Despite its strong empirical performance, RL-MCF has several limitations that warrant discussion. First, the current framework is restricted to univariate time series, as the autoencoder and actor-critic agent are designed for single-channel inputs. Extending the framework to multivariate settings introduces additional challenges, including the need to model inter-variable dependencies and to define actionability constraints across multiple channels simultaneously.

Second, the quality of the generated counterfactuals is sensitive to the hyperparameters controlling the target band, namely the gap parameter  $\rho$  and the band width  $fr$ . Poorly calibrated values may result in trivially easy or practically unreachable counterfactual objectives, directly affecting validity. While we provide default values that work well across our experimental settings, a principled hyperparameter selection strategy remains an open problem.

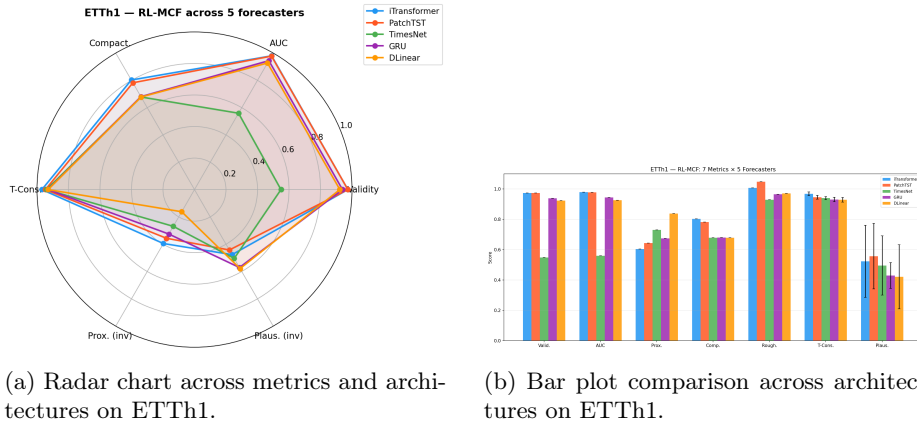


Fig. 4: Visual evaluation of RL-MCF on ETTh1 : comparison of five forecasting architectures across all metrics.

Third, the temporal masking mechanism assumes that actionable interventions are concentrated in the most recent timesteps, which may not hold in all application domains. In settings where relevant causal factors are distributed across the entire input horizon, the fixed ramp-based mask may unnecessarily restrict the agent’s expressiveness.

## 5 Ablation Study

We conduct two ablation experiments to validate the key components of our framework: the learned RL policy and the temporal masking mechanism. Table 3 summarizes the results.

Table 3: Ablation study: quantitative comparison of counterfactual quality metrics on the Transformer forecaster (ETTh1 dataset).

| Method          | Validity      | Step AUC      | Proximity    | Compactness   | Temp. Cons.   | Plausibility  |
|-----------------|---------------|---------------|--------------|---------------|---------------|---------------|
| RL-MCF          | <b>0.9833</b> | <b>0.9877</b> | <b>0.577</b> | <b>0.8448</b> | 0.93          | 0.5138        |
| RL-MCF w/o Mask | 0.8510        | 0.8552        | 2.059        | 0.0026        | 0.6427        | 0.45          |
| RLP             | 0.3544        | 0.3633        | 0.5857       | 0.8444        | <b>0.9628</b> | <b>0.4384</b> |

### 5.1 Effect of the Learned RL Policy (RLP)

To isolate the contribution of the learned RL policy, we introduce RLP as an ablation baseline that operates in the same latent space as our method, but replaces the actor-critic agent with random Gaussian noise:  $x_{cf} = \mathcal{D}(\mathcal{E}(x) + \epsilon)$ ,

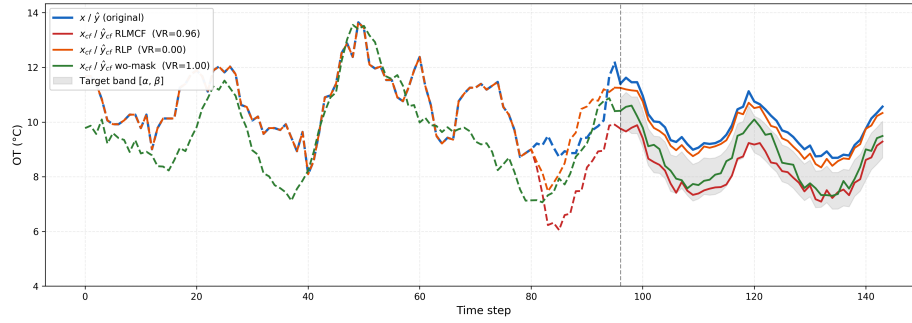


Fig. 5: Qualitative comparison of counterfactual time series generated by RLMCF, RLP (Random Latent Perturbation), and the wo-mask ablation on a representative test sample from ETTh1 (iTransformer forecaster).

where  $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ , with no reward signal or temporal masking applied. Without a learned policy, the agent fails to steer the forecast into the target band reliably (Validity: 0.3544 vs. 0.9833), confirming that the actor-critic policy is essential to balance proximity, compactness, and validity simultaneously. Although RLP achieves slightly higher temporal consistency (0.9628 vs. 0.93), this comes at the cost of dramatically lower validity, making the generated counterfactuals unreliable in practice.

## 5.2 Effect of Temporal Masking (Ours w/o Mask)

To assess the contribution of the temporal masking mechanism, we ablate it by disabling the mask, allowing the actor-critic agent to perturb all input timesteps freely without any sparsity constraint. All other components remain identical to our full method. Removing the mask causes a dramatic drop in compactness ( $0.8448 \rightarrow 0.0026$ ), confirming that the mask is solely responsible for enforcing sparse, localized perturbations. Validity also decreases ( $0.9833 \rightarrow 0.8510$ ) and proximity degrades significantly ( $0.577 \rightarrow 2.059$ ), indicating that unconstrained perturbations produce counterfactuals that deviate substantially from the original input. Both variants achieve high Step AUC, however the masked variant preserves compactness and proximity simultaneously, making the generated counterfactuals significantly more actionable and interpretable in practice.

## 6 Conclusion

In this paper, we proposed a reinforcement learning framework for counterfactual explanations in time series forecasting. By operating in the latent space of a pre-trained autoencoder, our actor-critic agent learns a global policy enabling counterfactual generation in a single forward pass, drastically reducing runtime compared to instance-specific gradient-based methods. The temporal masking mechanism further enforces sparse and localized perturbations, yielding compact and actionable explanations that target only the most recent and relevant timesteps. Extensive experiments on real-world datasets spanning energy and meteorology demonstrate that our method consistently achieves higher validity, better compactness, and shorter runtime than existing baselines, while producing counterfactuals that are realistic and directly actionable in practice. These properties make our framework particularly suited for high-stakes real-world deployment, where explanations must be not only valid but also efficient to compute and meaningful for end-users. Future work will extend the framework to multivariate settings and incorporate domain-specific actionability constraints.

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